

Bill of Materials (BOM) — 137.5 MHz VHF Satellite LNA Receiver

Quantity calculated for exactly **ONE (1)** single unit board. Optimized for R&D/Aerospace standards (High-Q, Tight Tolerance in RF Path).

Total components: 39 parts required for 1 fully populated board.

1. Semiconductors & Connectors

Qty	Part Number (MPN)	Package	Description
1	PGA-103+	SOT-89	High Dynamic Range MMIC LNA (U1)
1	AO3401A	SOT-23	P-MOSFET, Reverse Polarity Protection (Q1)
2	132289	SMD Edge	SMA Female Edge Launch Connector 50 Ω (J1, J2)
1	S2B-XH-A-1	TH 2.5mm	Header 2-Pin 2.5mm Right Angle (J3)

2. Inductors & Ferrite Beads (High-Q SMD)

Qty	Part Number (MPN)	Package	Value / Description
2	0603CS-16NXGRW	0603	16 nH, 2% , Ceramic Core (Butterworth)
1	0603CS-22NXGRW	0603	22 nH, 2% , Ceramic Core (Cauer)
2	0603CS-33NXGRW	0603	33 nH, 2% , Ceramic Core (Cauer)
1	0603CS-82NXGRW	0603	82 nH, 2% , Ceramic Core (Cauer)
1	0603CS-R12XGRW	0603	120 nH, 2% , Ceramic Core (Cauer)
1	0603CS-R15XGRW	0603	150 nH, 2% , Ceramic Core (Cauer)
1	0603CS-R22XGRW	0603	220 nH, 2% , Ceramic Core (Cauer)
1	0603CS-R27XGRW	0603	270 nH, 2% , Ceramic Core (Butterworth)
2	0805LS-471XJRC	0805	470 nH, 5% , Ceramic-Ferrite Core (Bias choke / R+L stability shunt) ¹
1	0805LS-681XJRC	0805	680 nH, 5% , Ceramic-Ferrite Core (Stab. network) ¹
1	BLM18HE601SN1D	0603	600 Ω @ 100MHz Ferrite Bead (DC Filtering)

¹ Kept at 5% intentionally: these set wideband RF-choke / stability-shunt impedance, not a precision frequency response. The design already carries margin (K-factor target 1.2-1.5, achieved 1.68), so $\pm 5\%$ has no meaningful effect. A 2% option exists in the 0805LS series but isn't needed here.

3. Capacitors (C0G/NP0 & X7R/X5R)

Qty	Part Number (MPN)	Package	Value / Description
1	GRM1885C1H4R0BA01D	0603	4 pF, ± 0.1 pF, 50V (C0G/NP0)
1	GRM1885C1H120GA01D	0603	12 pF, 2%, 50V (C0G/NP0)
2	GRM1885C1H220GA01D	0603	22 pF, 2%, 50V (C0G/NP0)
2	GRM1885C1H680GA01D	0603	68 pF, 2%, 50V (C0G/NP0)
1	GRM1885C1H820GA01D	0603	82 pF, 2%, 50V (C0G/NP0)
2	GRM1885C1H101GA01D	0603	100 pF, 2%, 50V (C0G/NP0)
1	GRM1885C1H151GA01D	0603	150 pF, 2%, 50V (C0G/NP0)
1	GRM1885C1H331JA01D	0603	330 pF, 5%, 50V (C0G/NP0) — Stab. network ¹
2	GRM188R72A103KA01D	0603	10 nF, 10%, 100V (X7R) — DC Block
1	GRM188R71C334KA01D	0603	0.33 μ F, 10%, 16V (X7R) — Decoupling
1	GRM188R60J106ME47D	0603	10 μ F, 20%, 6.3V (X5R) — Decoupling

4. Resistors (Thick Film, 1%)

Qty	Part Number (MPN)	Package	Value / Description
1	RC0603FR-0718RL	0603	18 Ω , 1/10W, 1%
1	RC0603FR-0750RL	0603	50 Ω , 1/10W, 1%
1	RC0603FR-07150RL	0603	150 Ω , 1/10W, 1%
2	RC0603FR-07300RL	0603	300 Ω , 1/10W, 1%